Paper Outline Draft

Working title candidates inside — IGLOO / Path-2 investigation

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Positioning: what's already published vs. what's ours

A literature pass turned up close cousins — good news, it means the niche is active and citable; the gap they leave is exactly where this paper sits.

Closest existing work:

- **"Residue-Level Attributions in Protein Language Models Do Not Recover Allergen Epitopes"** (arXiv 2606.22181, 2026) — same *symptom* (pLM attribution doesn't recover known-important residues), but on real biological annotations (noisy, incomplete ground truth) and *without* a mechanistic diagnosis or a fix. This is the paper reviewers will ask "how is yours different from this."
- **"Towards the Explainability of Protein Language Models"** (*Nature Machine Intelligence*, arXiv 2506.19532) — broad review/framework; good context-setting citation, not a competitor.
- General finding echoed across several 2026 papers: *"attention weights are not necessarily faithful causal explanations,"* and a documented *"dissociation between predictive performance, model faithfulness, and [domain] faithfulness"* in pLM classifiers — this is the same phenomenon we found, described elsewhere without the mechanism. Good external corroboration that we're not looking at a fluke of our synthetic setup.
- **Adebayo et al., "Sanity Checks for Saliency Maps"** (NeurIPS 2018) — cascading / model-parameter randomization test, image-classification only as far as the literature shows. We are the ones extending it to a pLM+regression-head setting — cite explicitly, frame cascading randomization as "we apply Adebayo et al.'s test to a new domain," not as a novel test.
- **Genomics ISM/gradient-correction literature** ("Correcting gradient-based interpretations of deep neural networks for genomics," *Genome Biology*; fastISM; compressed-sensing ISM) — establishes ISM as a standard, respected comparator (not something we invented) and documents the same gradient-saturation family of concerns in a sibling domain (regulatory genomics DNNs). Strong parallel-domain citation.
- Sundararajan et al. 2017 (IG), Shrikumar et al. 2017 (DeepLIFT / grad×input) — foundational method citations, standard.

What's actually new here, that the closest cousin (the epitope paper) doesn't have:

1. A **synthetic benchmark with exact, generator-defined ground truth** — lets us measure faithfulness directly instead of against noisy real annotations.
2. A **mechanistic explanation of *why*** attribution fails — the chain-rule argument plus three independent verifications (mechanism check, cascading randomization, factor ablation) that converge on one cause.
3. A **validated architectural fix** that simultaneously improves accuracy and attribution quality — not just a diagnosis, a prescription.
4. A **metric-reliability audit** — which evaluation metrics are trustworthy for this kind of study, independently useful to anyone building a similar benchmark.

Working titles (pick one, or riff)

1. *"Shallow Bottlenecks, Blind Gradients: Why Integrated Gradients Misattributes Protein Fitness Predictions — and How to Fix It"*
2. *"When the Pool is the Problem: Diagnosing and Repairing Attention-Pooling Bot-*

tlenecks in pLM-Based Attribution"

3. *"Does Your Attention Head Even See the Mutation? A Mechanistic Audit of Integrated Gradients for Protein Fitness Models"*
4. *"Faithful by Design: An Architectural Fix for Gradient Attribution on Frozen Protein Language Model Embeddings"* ← leans into the "fix," probably the strongest for a methods-focused venue

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Abstract (draft)

Integrated Gradients (IG) is the default tool for asking a trained protein fitness model *which residues matter*, but whether its answer reflects what the model actually learned — rather than the geometry of the frozen protein language model (pLM) embedding it was computed on — has not been systematically tested. Using synthetic fitness landscapes with exact, generator-defined ground-truth residue importance, we show that IG on the standard architecture (ESM-C embedding → shallow attention pool → deep MLP) correlates only weakly and inconsistently with model quality. We trace this to a structural cause: because every learnable nonlinearity sits *after* the attention pool, the chain rule forces $\partial F/\partial x_i \approx \mathrm{attn}_i \cdot g$ — a single, position-independent vector g scaled by a shallow, under-constrained attention weight — making the deep network's learned function invisible to the per-position gradient. We confirm this mechanism three independent ways (direct verification, Adebayo-style cascading parameter randomization, and factor ablation of the IG product), and show a simple architectural fix — moving the deep MLP to run per-residue, *before* pooling — simultaneously improves predictive accuracy (Spearman ρ 0.68→0.78) and restores IG's ability to track model quality (discrimination r +0.41→+0.84), corroborated independently by a gradient-free perturbation baseline (ISM). We further show which of nine common attribution-evaluation metrics are actually trustworthy for this kind of audit, and two architectural changes that *don't* work (a vanishing-gradient failure and a capacity/data mismatch), clarifying that the fix is specific, not incidental.

(Numbers above are from the pilot-scale runs; update once the full 9-level sweep — currently rebuilding — completes.)

Section-by-section outline

1. Introduction

- Hook: protein engineers use fitness models not just for prediction but to *learn which residues matter* — active-site redesign, stability engineering, directed evolution prioritization. IG is the default tool for this.

- The question nobody had ground truth to answer: does IG's answer reflect the model, or the frozen pLM embedding underneath it?
- One-sentence preview of the finding + the fix.
- **Generalization hook** (put this in the intro, not just the discussion): *frozen-encoder + shallow-pool + trainable head* is now the default recipe far beyond protein ML — CLIP linear probes, LoRA-adapted heads, DNA-LM regressors. The mechanism here is a property of that architecture shape, not a protein-specific quirk.

2. Background / Related Work

- IG (Sundararajan et al. 2017), axioms (Sensitivity, Completeness).
- DeepLIFT / grad×input (Shrikumar et al. 2017).
- Adebayo et al. 2018 sanity checks — cite explicitly as the source of the cascading-randomization test; note it has not (to our knowledge) been applied to a pLM+regression setting.
- pLM interpretability: the epitope-recovery paper, the NMI explainability review, "attention is not [not] explanation" debate (Jain & Wallace / Wiegreffe & Pinter) — note ours is a gradient-based, not raw-attention, version of that concern.
- Genomics parallel: ISM as an established comparator (fastISM, compressed-sensing ISM), gradient-correction-for-saturation literature (Genome Biology) — same family of concerns in a sibling domain.

3. Synthetic benchmark

- GB1 (56 aa) and T4 Lysozyme (164 aa); additive / epistatic / nonlinear fitness landscapes with BLOSUM-weighted, category-annotated ground-truth importance (`binding_site` / `structural` / `folding` / `active_site` / `metal_binding`, etc.).
- Why synthetic: exact ground truth, no confound between "biologically correct" and "recovers the generator's rule."
- Label-noise sweep as the tool for producing models of controlled, known quality: `y := renorm((1-p)·y_true + p·y_perm)`, $p \in [0,1]$ scale-free (0 = exact labels, 1 = fully random) — cleaner than an absolute-σ Gaussian sweep (report the std-vs-blend comparison as a methods contribution; absolute σ conflates noise level with each landscape's own label scale and never actually reaches $\rho \approx 0$).

4. Baseline result: IG is a weak, inconsistent quality signal

- Report the original (bf16-era) finding, contrasted with the corrected finding once labels actually spanned the noise range (this progression is itself worth narrating briefly — it's a good methods lesson: too narrow a noise sweep manufactures a false negative).
- Figure: model test ρ (x) vs. IG spearman-to-ground-truth (y), colored by noise, baseline architecture only.

5. Mechanism

- Derive $\partial F/\partial x_i \approx \text{attn}_i \cdot g$ from the architecture (`ESMAttention1d`).
- **Verification 1 — direct check**: pearson(attn_i, mean$\|\text{IG}_i\|$) = 0.84–0.99; cosine dispersion of `g` across variants ≈ 0.002–0.07 (≈ one fixed vector).
- **Verification 2 — cascading randomization** (Adebayo-style): randomizing `final→fc4→fc3→fc2→fc1→layernorm` leaves attribution correlation at 0.85–1.0; only randomizing `attention1d` collapses it.
- **Verification 3 — factor ablation**: decompose the IG product into `gradient` (model-only) and `displacement` (pLM-geometry-only, ρ≈0.12 flat across every condition) factors; `gradient` alone ≈ the attention pattern.
- One figure per verification; these three converging independently is the strongest evidence in the paper — say so explicitly.

6. The fix: move depth before the pool

- `perres_mean` / `perres_attn` (entire MLP runs per-residue on the raw embedding, aggregated last by mean / learned attention) vs. `IG_premlp` (smaller residual per-residue layer before the existing pool).
- Result: accuracy **and** discrimination both improve (ρ 0.68→0.72–0.78; discrimination r +0.41→+0.84 for `perres_mean`, the strongest of any architecture tested).
- Mechanistic explanation for *why* it works: mean-pool has a uniform 1/L gradient, so 100% of the per-position IG signal is forced through the trained per-position MLP, not a shallow attention scalar.

7. It's not "any architecture change helps" — two negative results

- `channel_attn`: attention pools the channel axis instead of length — a **vanishing-gradient failure**, not a subtler version of the same problem. Show the raw-magnitude comparison (attributions ~10,000× smaller than baseline) and the exactly-identical noise-0-vs-noise-1 metric distributions. Root cause: two stacked softmax bottlenecks + collapsing every position to one scalar before any MLP runs.
- `crossattn`: adding real cross-position self-attention *hurts* accuracy at this data scale (0.48 vs. 0.68 baseline) — capacity/data mismatch, and its discrimination on hit-rate metrics is the weakest of the working architectures. Frame as: more expressive ≠ better, at this training-set size.
- Together these show the fix is *specific* (remove the post-pool bottleneck) rather than "architecture surgery helps in general."

8. Corroboration: a gradient-free method agrees

- ISM (in-silico mutagenesis) — no gradient, no embedding-space assumption, established in the genomics literature as a faithful (if expensive) comparator.
- ISM independently shows the same model-quality-tracking pattern (spearman r = 0.64–0.87 depending on landscape) — rules out "this is an IG-specific artifact" as an alternative explanation.

- Report the honest caveat found along the way: ISM's *sign*-based metric disagreed in direction from every gradient method (r = −0.44 on `sign_acc`) — plausibly a sign-convention mismatch (raw prediction delta vs. path-integrated gradient dot product), flagged as future work to resolve rather than papered over.

9. Which evaluation metrics can you trust?

- Nine metrics audited (spearman, auroc, precision@k, recall_topcat, topcat_rank, enrichment, category_mono, epi_ratio, sign_acc) against the full noise sweep, across every architecture.
- `spearman` and `sign_acc`: reliable, positive discrimination almost everywhere.
- `precision@k` / `recall_topcat`: real signal, but protein-prevalence-dependent chance floors (GB1's 77% annotation density vs. T4's 4%) make raw values incomparable across proteins — always skill-normalize.
- `category_mono`: indistinguishable from noise on average.
- `epi_ratio`: actively misleading (average discrimination flips sign once skill-normalized) *and* structurally undefined for densely-annotated proteins like GB1 (the "non-contact annotated" reference group is empty) — worth a explicit worked derivation of why, it's a nice self-contained methods point.
- Practical contribution: a reusable checklist for anyone building a similar ground-truth benchmark.

10. A striking random-baseline result

- Under full label noise (p=1), `sign_acc` doesn't fall to chance (0.5) — it drops to ≈0.19–0.32 (pooled), *below* chance: a model trained on pure noise doesn't just lose the correct sign, it systematically prefers the wrong one. Good, quotable result for the discussion of what "learned nothing" looks like at the attribution level.

11. Discussion

- Restate the generalization claim with the specific chain-rule condition spelled out: *any* architecture where 100% of the trainable nonlinear depth sits after a pooling/aggregation step will have this problem, regardless of domain.
- Practical recommendations for practitioners (ranked, from the "what would improve the attributions" analysis):
  1. Architecture: move depth before pooling.
  2. Cross-check gradient attribution against a perturbation method.
  3. Metric choice: spearman / sign_acc first; skill-normalize anything with a prevalence-dependent floor.
  4. Sanity-check any new architecture with cascading randomization before trusting its attributions.
- Limitations (see gaps below) stated plainly, not buried.

12. Limitations & Future Work

- **Everything is on synthetic data** — the single biggest limitation; state it early and directly.
- Token-level IG (interpolating through the pLM's own weights, not just the small head) is mathematically informative only for multi-mutants — derive briefly, point to it as a partially-explored future direction.
- Causal mediation / activation patching through the pLM as the correct tool for testing whether the model-independent "displacement" signal (ρ≈0.12) has exploitable structure — not yet built, name it as the clear next step.
- ISM sign-convention discrepancy unresolved.

Figures checklist (mapped to what already exists)

#	Figure	Source
1	Model quality vs. IG quality, baseline arch	`results/synthetic/metric_suite/metri`
2	Mechanism verification scatter	`results/synthetic/attribution_mechan`
3	Cascading randomization curves	`results/synthetic/cascading_randomi`
4	Factor ablation decomposition	`results/synthetic/factor_ablation_an`
5	Architecture leaderboard (accuracy)	"The Attribution Trial" artifact, Q1
6	Discrimination heatmap, all architectures	"The Attribution Trial" artifact, Q2/Q3
7	`channel_attn` degeneracy illustration	"The Attribution Trial" artifact, Q3
8	ISM vs. IG agreement	`results/synthetic/ism_attribution/`
9	Metric usefulness ranking	"The Attribution Trial" artifact, Q5
10	sign_acc collapse below chance at full noise	"The Attribution Trial" artifact, Q4

Still needed: one qualitative vignette (a specific mutation, baseline vs. fixed architecture, tied to a named annotated site) — none of the current figures show a single concrete example, only aggregates.

Gaps before this is submission-ready

1. **Real (non-synthetic) validation case** — reviewers will not accept synthetic-only; even one small real dataset with literature-known catalytic/binding residues as a sanity check matters a lot for credibility.
2. **Finish the full 9-level noise sweep** on the winning architectures (currently rebuilding after the BOS/EOS embedding fix) — headline numbers should come from the full sweep, not the 2-point pilot.
3. **Dose-response ablation**: does one per-residue layer suffice, or does more depth help further? Currently one data point per architecture.
4. **The qualitative vignette** (see figures checklist).
5. Resolve or at least characterize the ISM sign-convention discrepancy before presenting it as "corroboration" — right now it's corroboration-with-an-asterisk.
6. Decide on venue (see below) — affects framing emphasis and length.

Possible venues

- **ISMB / Bioinformatics / PLOS Computational Biology** — biology-forward framing (Sections 1–3, 6, 9 lead); mechanism section can be more compact.
- **NeurIPS / ICML interpretability workshop, or the main track as an application paper** — mechanism-forward framing (Section 5 leads); the generalization-beyond-protein argument needs to be earned with at least one non-protein toy example if aiming here.
- **A short technical note to EvolutionaryScale / an ESM-C GitHub issue**, separate from the paper — the BOS/EOS special-token bug in `ESMCBackend`-style high-level embedding APIs is a distinct, self-contained, practically useful finding (anyone using ESM-C's `.logits(..., return_embeddings=True)` for per-residue analysis silently gets an off-by-one). Worth reporting independently regardless of the paper's timeline — it's not really "our" finding to sit on if it affects other users right now.